HW11

December 10, 2025

Exercise 1 (Conservation law) Assume that $H(0) = 0$ and $u \in \mathcal{C}_c([0,\infty) \times \mathbb{R})$ is a weak solution to

$$u_t + \big(H(u)\big)_x = 0, \quad u(0,x) = g(x),$$

that is, for every $v \in \mathcal{C}_c^\infty([0,\infty) \times \mathbb{R})$,

$$\int_{\mathbb{R}} dx \int_0^\infty dt \left[u v_t + H(u) v_x \right] + \int_{\mathbb{R}} g(x) v(0,x)\, dx = 0.$$

Show that

$$\int_{-\infty}^{\infty} u(\cdot, t)\, dx = \text{constant},$$

and identify the constant.

Hint: take $v(t,x) = \phi(t)\chi(x)$ where $\chi(x) \approx \mathbb{1}_{[-R,R]}(x)$ for sufficiently large R.

Exercise 2

1. Use the Hopf–Lax formula to solve the Burgers equation

$$u_t + u \cdot u_x = 0, \quad u(0,x) = g(x),$$

with the initial condition

$$g(x) = \begin{cases} 1, & x < -1, \\ 0, & -1 < x < 0, \\ 2, & 0 < x < 1, \\ 0, & x > 1. \end{cases}$$

2. Draw a picture of the characteristics and verify the Rankine–Hugoniot condition along the shocks.

Ex1. 取 $v(t,x)=\phi(t)\chi(x)$

代入弱解定义式得 $\int_{\mathbb{R}}dx\int_0^\infty[u\phi'(t)\chi(x)+H(u)\phi(t)\chi'(x)]dt+\int_{\mathbb{R}}g(x)\phi(0)\chi(x)dx=0$

对于 $\forall t\geq 0$，取充分大的 R，使 $u(t,\cdot)$ 的紧支集包含在 $[-R,R]$ 中

此时在支集上 $\chi\equiv 1$，即 $\chi'\equiv 0$

从而上式化简为 $\int_0^\infty\phi'(t)dt\int_{\mathbb{R}}u(t,x)dx+\phi(0)\int_{\mathbb{R}}g(x)dx=0$

对于 ϕ s.t. $\phi(0)=0$，由上式得 $\int_0^\infty\phi'(t)dt\int_{\mathbb{R}}u(t,x)dx=0$，

即在分布意义下 $\frac{d}{dt}\int_{\mathbb{R}}u(t,x)dx=0$，即 $\forall t>0$，$\int_{\mathbb{R}}u(t,x)dx\equiv C$ 常数.

从而对于 ϕ s.t. $\phi(0)\in\mathbb{R}$，$-C\phi(0)+\phi(0)\int_{\mathbb{R}}g(x)dx=0$

即 $\int_{\mathbb{R}}u(t,x)dx=C=\int_{\mathbb{R}}g(x)dx$.

Ex2. (1) 对应的 Hamilton-Jacobi 方程为 $v_t+\frac{1}{2}v_x^2=0$，$v(0,x)=\int_0^x g(s)ds$

注意到，$v(0,x)=\begin{cases}x+1, & x<-1\\ 0, & -1\leq x<0\\ 2x, & 0\leq x<1\\ 2, & x\geq 1\end{cases}$，$H(p)=\frac{1}{2}|p|^2$，$L(p)=\frac{1}{2}|p|^2$

故由 Hopf-Lax formula，$v(t,x)=\min_y\left\{\frac{1}{2t}(x-y)^2+v(0,y)\right\}$

从而 $u(t,x)=v_x(t,x)$

当 $0<t<1$，$u(t,x)=\begin{cases}1, & x<-1+\frac{1}{2}t\\ 0, & -1+\frac{1}{2}t<x<0\\ \frac{x}{t}, & 0<x<2t\\ 2, & 2t<x<1+t\\ 0, & x>1+t\end{cases}$；当 $1\leq t<2$，$u(t,x)=\begin{cases}1, & x<-1+\frac{1}{2}t\\ 0, & -1+\frac{1}{2}t<x<0\\ \frac{x}{t}, & 0<x<2\sqrt{t}\\ 0, & x>2\sqrt{t}\end{cases}$；

当 $2\leq t<6+4\sqrt{2}$，$u(t,x)=\begin{cases}1, & x<t-\sqrt{2t}\\ \frac{x}{t}, & t-\sqrt{2t}<x<2\sqrt{t}\\ 0, & x>2\sqrt{t}\end{cases}$；当 $t\geq 6+4\sqrt{2}$，$u(t,x)=\begin{cases}1, & x<1+\frac{1}{2}t\\ 0, & x>1+\frac{1}{2}t\end{cases}$

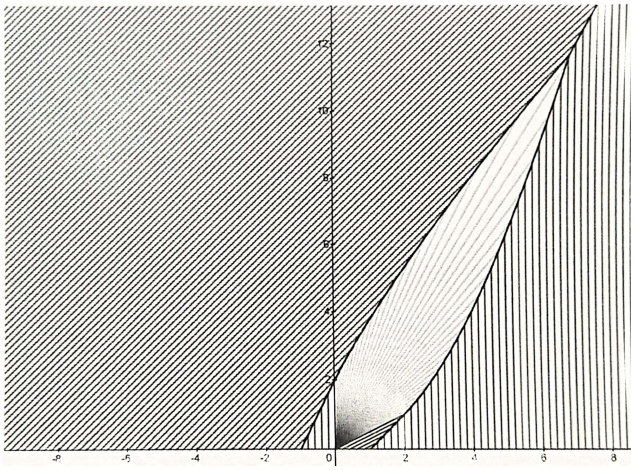
(2) 特征线 $x=x_0+g(x_0)t$.

Shocks: $S_1: x=\begin{cases}-1+\frac{1}{2}t, & 0\leq t\leq 2\\ t-\sqrt{2t}, & t\geq 2\end{cases}$

$\dot{S}_1=\frac{1}{2}$，$\frac{u_L+u_R}{2}=\frac{1+0}{2}=\frac{1}{2}$，✓

$\dot{S}_1=1-\frac{\sqrt{2}}{2\sqrt{t}}$，$\frac{u_L+u_R}{2}=\frac{[1-\frac{\sqrt{2}}{\sqrt{t}}]+[1-\frac{1}{\sqrt{2t}}]}{2}=1-\frac{\sqrt{2}}{2\sqrt{t}}$，✓

$S_2: x=\begin{cases}1+t, & 0\leq t\leq 1\\ 2\sqrt{t}, & t\geq 1\end{cases}$

$\dot{S}_2=1$，$\frac{u_L+u_R}{2}=\frac{2+0}{2}=1$，✓

$\dot{S}_2=\frac{1}{\sqrt{t}}$，$\frac{u_L+u_R}{2}=\frac{\frac{2}{\sqrt{t}}+0}{2}=\frac{1}{\sqrt{t}}$，✓